

AI / Embodied Intelligence / Embedded Systems

Zhang Quanrui

BUPT / Telecommunications Engineering with Management / Class of 2029

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Study Room

Profile

BUPT undergraduate with a first-year weighted average of 90.85/100 (top 5%). Experience spans leading a three-person AIC autonomous-vehicle team on ROS navigation and mission logic, PRIS Lab embodied AI training, independent time-series forecasting, embedded robot control, and autonomous picking simulation.

Education

BUPT / Telecommunications Engineering with Management — Class of 2029

B.Eng. Candidate / Sep 2025 - Jun 2029 (Expected)

Weighted average 90.85/100 / Top 5% / CET-4 530

- **Academics:** First-year weighted academic average of 90.85/100, ranking in the top 5% of the major.
- **Honors:** BUPT First-Class Scholarship, 2025-2026 academic year.
- **Coursework:** Advanced Mathematics I (88), Advanced Mathematics II (89), Linear Algebra (90), Introduction to Computing and Programming (91), Introduction to Computing and Programming Lab (99), and Fundamentals of Electronic Systems (98).
- **English:** College English Test Band 4 (CET-4): 530.

Research Experience

Pattern Recognition and Intelligent Systems Lab (PRIS Lab) — Ongoing

Undergraduate Researcher / Advisor: Prof. Gao Sheng

Embodied AI reading / Blueberry-picking robot / System integration

- **Focus:** Develop a connected understanding of agents and embodied intelligence, from model mechanisms to robotic system constraints.
- **Outcome:** Applied the lab training in a blueberry-picking robot competition project, covering environment setup, serial and teleoperation development, vision-control integration, and mobile-platform testing.

Short-term Bike-sharing Demand Forecasting with LSTM — Experimental report completed

Independent Experiment, Analysis, and Report

Python / LSTM / 24-hour input / 8-hour forecast

- **Problem:** Forecasted the next eight hours of bike demand from the preceding 24 hours, using chronological training, validation, and test splits.
- **Method:** Processed about 100,000 Shanghai Mobike orders into 713 hourly samples for imputation, normalization, and a two-layer LSTM; later evaluated about one million Shenzhen open-data orders with cyclical, weekend, and peak-hour features.
- **Results and limits:** Initial tuning reduced MAE from 77.82 to 61.45 and RMSE from 105.26 to 91.61. The expanded-data experiment reached an R^2 of roughly 0.3-0.4; sparse spikes and a 28.6% zero-value share were documented as major limitations.

BUPT-Tsinghua Embodied Intelligence Knowledge Base — Internal knowledge base

Core Contributor / University Embodied Intelligence Alliance

124 scholars / 122 labs / Structured academic resources

- **Role:** Owned information collection, manual verification, and structured entry for the Core Scholars and Core Labs modules.
- **Scale:** Completed structured records for 124 scholars and 122 laboratories, supporting a searchable and extensible academic resource base.



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Engineering Projects

AIC Autonomous Vehicle Visual Cruise: Navigation and Mission System — **Competing; results pending**

Team Lead (3-person team) / Navigation, Mission Logic, and Testing

ROS Melodic / Python / AMCL + move_base / 2D LiDAR / VLM / Offline simulation

- **Architecture:** Designed the mission ROS package on the vendor AMCL/move_base stack: a state machine chaining wake-up, recognition, navigation, box entry, and speech; an online LiDAR baffle grid map with cell-level routing; box entry with clearance from the nearest measured baffle point, in two stages when a baffle faces the box; and a time guard from remaining distance and measured speed. Teammates wrote the first recognition and TTS versions; I integrated and optimized them.
- **Root-cause debugging:** Ran a log-to-retest loop: traced corridor stalls to a global inflation setting that cut a 43 cm corridor's plannable width from 23 to 11 cm (fix: planning failures 525 to 80, run time 222 to 167 s); a one-day review of 18 runs traced 8 baffle contacts to 4 root causes, such as 2 cm optimistic ranging and tail swing in tight turns, each addressed; disabling VLM deep thinking cut recognition from 15-54 s to 2-3 s.
- **On-car results:** Over 18 on-car runs on 18-baffle layouts in one day, shipped five successive versions and raised the strictly team-graded average from 64 to 87 (out of 100), with a clean run reaching all four boxes; 69 of 70 clue reads were correct, and angled shots cut the slowest wall from 29-75 s to 17-18 s.

Blueberry-picking Robot: Embedded Control and Picking Simulation — **Simulation delivered**

Primary Developer / Embedded Control, Integration, and Delivery

Renesas RA8P1 / Python / YOLO11n / Blender / MuJoCo / Panda

- **Embedded and hardware:** Developed RA8P1 serial communication and arm teleoperation; integrated an existing YOLO detector into control commands and tested the arm on a mobile chassis.
- **Autonomous simulation:** Contributed to a Blender/YOLO11n/MuJoCo/Panda loop using localization, inverse kinematics, state control, and contact feedback for grasping, shrub picking, transfer, and tray verification.
- **Simulation evaluation:** On frozen simulation scenes, achieved 30/30 successful tabletop grasps, 6/6 correct no-target refusals, and 17/20 successful simplified shrub trials, with no unripe targets picked.

NaviGoose Assistive Sensing System for the Visually Impaired — **Nestling Plan project completed**

AI Scene-recognition Design / Advisor: Assoc. Prof. Zhang Yang

LiDAR / 77GHz mmWave / Edge-cloud architecture

- **Contribution:** In a five-person team, owned AI scene-recognition planning and architecture analysis, and independently prepared the technical proposal, competitor research, and defense materials.

Machine Learning Portfolio

Machine Learning and Deep Learning Notes: Mechanism-first notes with formulas, runnable practice, and review checks.

ML fundamentals, Neural networks and deep learning, Self-Attention, Transformer, BERT, Practice and review / Public curated edition

Skills and Tools

Programming and Data	C/C++, Python; NumPy, Pandas, PyTorch, and Scikit-learn.
Models and Algorithms	LSTM, Transformers, and attention mechanisms; continuing study of VLA models and embodied intelligence.
Robotics and Embedded	ROS Melodic, AMCL and move_base (DWA) tuning, 2D LiDAR geometry, Linux onboard deployment; Renesas RA8P1, e2 studio, serial communication, robotic-arm teleoperation, YOLO inference integration, and hardware testing.
Research and Development Practice	Literature search, paper reading, and experiment logging; Zotero, Anaconda, Markdown, plus AI-assisted programming, debugging, and technical documentation.

Selected Awards and Activities

- CSP-J Olympiad in Informatics / Provincial Second Prize
- Lab4AI × BUPT AI-powered Academic Writing Workshop / Completion Certificate
- BUPT 2025 Nestling Plan / Contributor to a Completed Project
- International School Freshman Debate Tournament / Top 6